INDIA'S ENERGY CONSUMPTION TRENDS

A Decadal Analysis of Growth, Sectoral Demand, Energy Mix, Emissions and Policy Outcomes (FY 2013-14 to FY 2022-23)

A Data-Driven Report Based on the Energy Statistics India 2024 Publication

(Ministry of Statistics and Programme Implementation, Government of India)

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Executive Summary

India's total primary energy consumption rose from 26,822 petajoules (PJ) in FY 2013-14 to 35,159 PJ in FY 2022-23(P) — a compound annual growth rate (CAGR) of 3.05% over the decade, briefly interrupted by a 8.4% contraction in FY 2020-21 due to the COVID-19 pandemic. Coal remains the backbone of the energy system, supplying 58.98% of total primary energy supply (TPES) in FY 2022-23, followed by crude oil (31.48%) and natural gas (6.61%); nuclear and renewables together contribute only about 5.2%. Fossil fuels therefore still account for roughly 97% of TPES, even though renewable electricity capacity (excluding large hydro) has grown at a much faster 15.21% CAGR — from about 35 GW in FY 2013-14 to 125 GW in FY 2022-23 — outpacing overall generation capacity growth of 5.89% CAGR.

Sectorally, industry is the largest consumer of electricity (42.4% share in FY 2022-23), followed by the domestic/residential sector (25.79%), agriculture (17.16%) and commercial establishments (7.49%); the domestic sector has grown fastest (6.82% CAGR) on the back of rising electrification and appliance ownership. India's per-capita energy consumption, at roughly 25,424 megajoules (about 25-27 gigajoules) in FY 2022-23, remains far below China (~120 GJ), the United States (~277 GJ) and the world average (~77 GJ), reflecting India's early stage of industrialisation and still-developing energy access rather than efficient use alone. Government programmes — the National Solar Mission, UJALA, and the Perform, Achieve and Trade (PAT) scheme — have driven measurable gains: solar tariffs have fallen roughly 65% since 2014-15, over 36 crore LED bulbs have been distributed, and India met its 50% non-fossil capacity target five years ahead of the 2030 deadline. Yet import dependency for crude oil and natural gas has actually risen over the decade, and coal consumption continues to grow in absolute terms, underscoring that India's transition is additive (renewables layered on top of growing fossil use) rather than substitutive at this stage. The report closes with a review of projected demand growth to 2035-2047 and policy recommendations for a more secure and sustainable energy pathway.

1. Total Primary Energy Consumption: Trends, Sources and Growth Drivers (2013-14 to 2022-23)

1.1 Annual Consumption Trend

India's total primary energy consumption grew steadily from 26,822 PJ in FY 2013-14 to a peak pre-pandemic level of 32,548 PJ in FY 2019-20, before falling to 29,807 PJ in FY 2020-21 (a decline of 8.4%) as COVID-19 lockdowns suppressed industrial and transport activity. Consumption rebounded strongly thereafter, reaching 33,018 PJ in FY 2021-22 and 35,159 PJ in FY 2022-23(P) — a year-on-year increase of 6.48%. Over the full decade the CAGR works out to 3.05% per year.

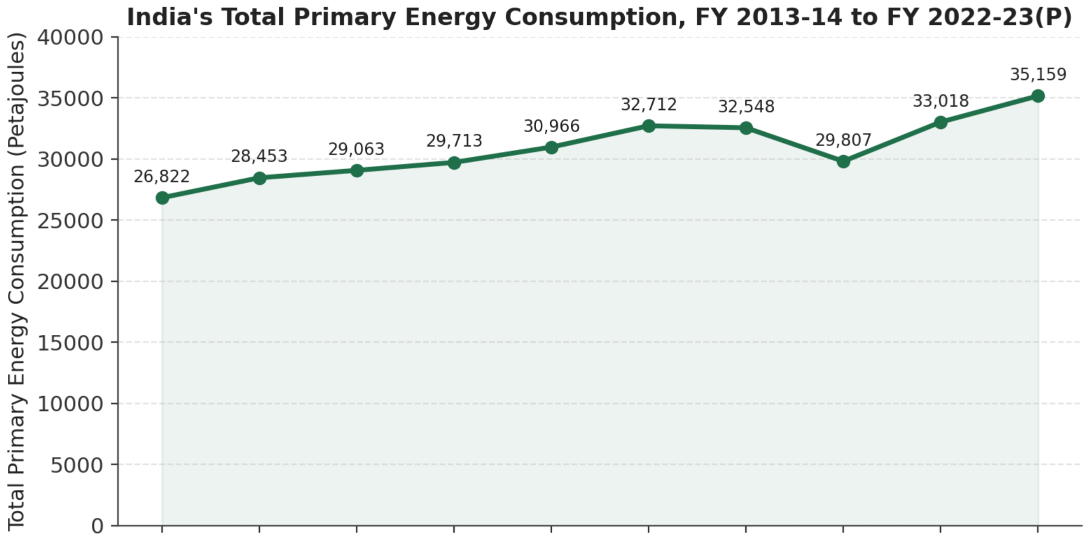

Figure 1: Total Primary Energy Consumption, FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 6.2.

1.2 Source-wise Contribution to Growth

Coal and lignite together dominate the energy mix, and coal alone drove most of the absolute growth in consumption over the decade — rising from 14,214 PJ to 20,257 PJ (CAGR 4.02%). Crude oil consumption grew more slowly (CAGR 1.54%, from 9,520 to 10,921 PJ) and natural gas grew at a similar pace (CAGR 1.58%). The 'electricity' component of Table 6.2 — which captures the energy embedded in hydro, nuclear and renewable generation after transmission losses — grew fastest of all conventional categories, at a 7.12% CAGR (652 PJ to 1,211 PJ), nearly doubling over the decade.

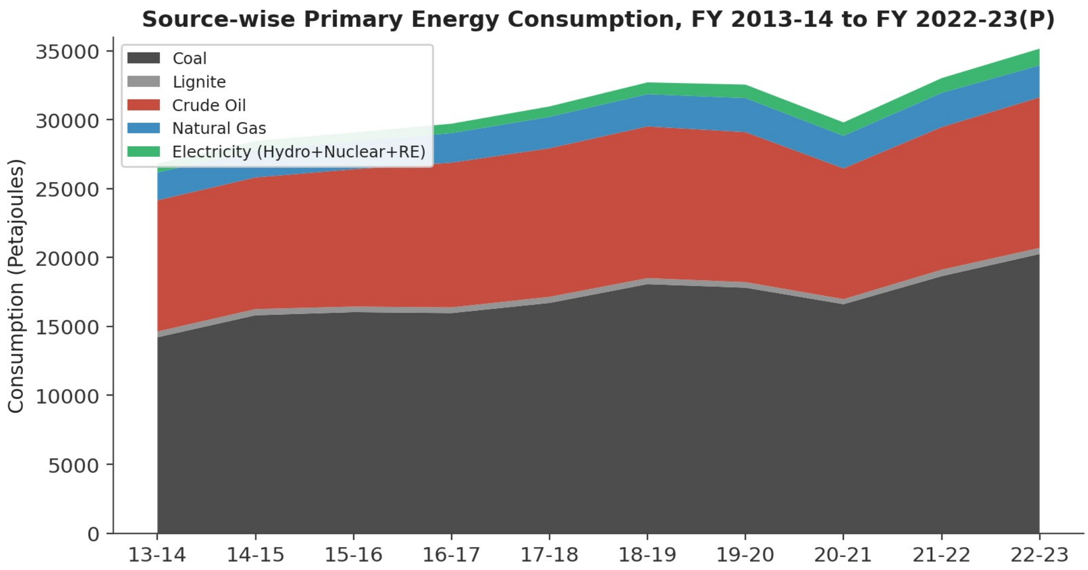

Figure 2: Source-wise Primary Energy Consumption (stacked), FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 6.2.

Source	2013-14 (PJ)	2022-23(P) (PJ)	Share in 2022-23 (%)	CAGR 2013-14 to 2022-23 (%)
Coal	14,214	20,257	57.6	4.02
Lignite	419	447	1.3	0.72
Crude Oil	9,520	10,921	31.1	1.54
Natural Gas	2,017	2,323	6.6	1.58
Electricity (Hydro+Nuclear+RE)*	652	1,211	3.4	7.12
Total	26,822	35,159	100.0	3.05

**Electricity here reflects hydro, nuclear and other renewable generation from utilities, net of transmission losses — it is a partial, energy-accounting proxy for non-fossil electricity rather than a complete renewables figure. Source: Energy Statistics India 2024, Table 6.2.*

1.3 Conventional vs. Renewable Energy: Comparing Growth Rates

Measured on a primary-energy basis, conventional fuels (coal, oil, gas) still dwarf renewables in absolute terms, but renewables are growing several times faster on every metric available in the report:

- Installed renewable capacity (RES, excluding large hydro) grew from about 35 GW in FY 2013-14 to 125 GW in FY 2022-23 — a CAGR of 15.21%, compared with just 4.28% CAGR for coal/steam thermal capacity and 5.89% CAGR for total installed capacity across all sources.
- The renewable share of total electricity generation rose steadily from 17.22% in FY 2013-14 to 20.34% in FY 2022-23, even as absolute generation from thermal sources kept growing.
- On a primary-energy-supply basis, however, the combined share of nuclear plus renewables in TPES inched up only marginally — from about 4.5% in FY 2013-14 to 5.2% in FY 2022-23 — because total energy demand itself is growing quickly and coal continues to be the preferred source for new baseload capacity.

In short: renewables are the fastest-growing part of India's energy system by capacity and by generation share, but conventional fuels are still capturing the largest share of the growth in absolute energy terms — the transition so far is best described as additive rather than substitutive.

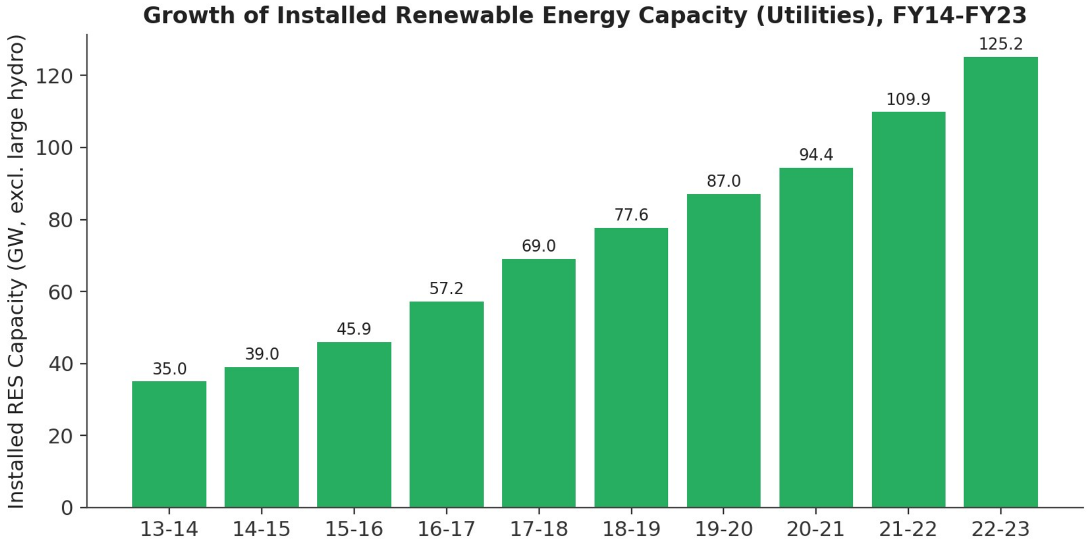

Figure 3: Growth of Installed Renewable Energy Capacity (utilities, excluding large hydro), FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 2.3(A).

1.4 Major Factors Driving Changes in Energy Demand

- Economic growth: India's GDP (at 2011-12 prices) grew at a 5.65% CAGR over the decade, directly pulling up energy demand for industry, construction, and freight/passenger transport.
- Population and urbanisation: mid-year population grew at 1.11% CAGR; the report separately notes that rapid urban growth is adding housing, commercial floor space and infrastructure that need power, cooling and mobility.
- Electrification and appliance penetration: near-universal household electrification plus rising ownership of fans, air-conditioners, refrigerators and other appliances has driven the domestic sector's above-average 6.82% CAGR in electricity consumption.
- Industrialisation: energy-intensive industries such as steel, cement and sponge iron are large and growing consumers of coal and electricity, keeping coal demand on an upward trajectory.
- Vehicle ownership and freight growth: consumption of High-Speed Diesel Oil (the single largest petroleum product by volume, ~38.5% of the total) is dominated by the transport sector and has grown accordingly.
- The COVID-19 shock and rebound: the sharp FY 2020-21 dip and the subsequent 6-7% annual rebounds illustrate how closely energy demand tracks the pace of overall economic activity.

2. Sector-wise Energy Consumption

2.1 Comparing Sector-wise Consumption

Because different fuels serve different end-uses, sector-wise patterns are best examined fuel-by-fuel. The most complete sectoral breakdown available is for electricity (Table 6.8), which covers industry, domestic, agriculture, commercial, traction/railways and 'others':

Sector	2013-14 (GWh)	2022-23(P) (GWh)	Share in 2022-23 (%)	CAGR (%)
Industry	3,84,418	5,95,000	42.40	4.97
Domestic	1,99,842	3,62,000	25.79	6.82
Agriculture	1,52,744	2,40,800	17.16	5.19
Commercial	74,247	1,05,100	7.49	3.94
Traction & Railways	15,540	25,000	1.78	5.43
Others	47,418	75,500	5.38	5.30
Total	8,74,209	14,03,400	100.00	5.40

Source: Energy Statistics India 2024, Table 6.8.

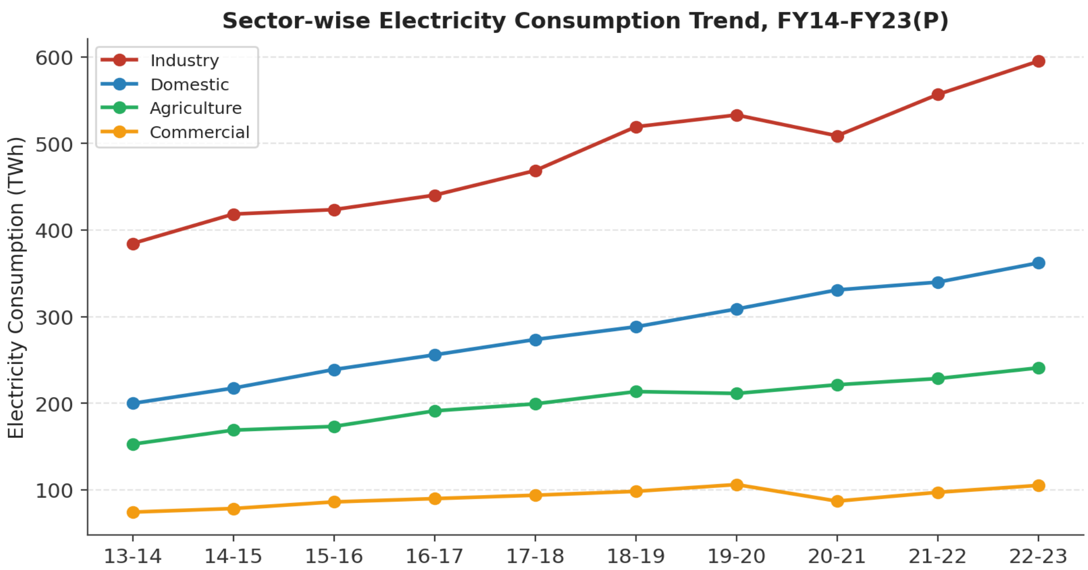

Figure 4: Sector-wise Electricity Consumption Trend, FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 6.8.

For coal, the electricity/power sector is overwhelmingly dominant, consuming 70.4% of all coal used in FY 2022-23, followed by imported coking coal for steel and washeries (6.35%) and cement (0.73%). For natural gas, the largest use is the fertiliser industry (32.35% of total consumption), followed by city/local gas distribution networks including CNG for road transport (20.06%) and power generation (13.59%). For petroleum products, transport is the dominant end-use — High-Speed Diesel Oil, the single largest product by volume, is consumed overwhelmingly by the transport sector, and its transport-linked consumption rose from about 71,700 thousand tonnes in FY 2018-19 to 80,026 thousand tonnes in FY 2022-23.

2.2 Why Certain Sectors Dominate Energy Demand

- Industry leads electricity demand because India's growth model still leans on energy-intensive manufacturing — steel, cement, textiles, fertilisers and chemicals — all of which have high specific energy consumption per unit of output.

- The domestic sector's rapid growth (highest electricity CAGR at 6.82%) reflects near-complete village/household electrification achieved over the last decade, combined with rising incomes that translate directly into appliance ownership (fans, ACs, refrigerators, televisions).
- Agriculture's large and rising electricity share is driven by groundwater-irrigation pump-sets, which are numerous, often subsidised, and used intensively during the cropping seasons.
- Coal's dominance in the power sector reflects India's continued reliance on coal-fired thermal plants for reliable, low-cost baseload power, even as renewable capacity expands.
- Transport's dominance in diesel/petrol demand reflects rapid growth in vehicle ownership, e-commerce-driven freight movement, and limited penetration of electric mobility to date.

2.3 Impact of Urbanisation, Industrialisation and Population Growth

These three structural forces reinforce one another. Population growth (1.11% CAGR) sets a rising floor for total demand across every end-use. Urbanisation concentrates that population into cities with more energy-intensive lifestyles — more appliances, more air-conditioning, denser transport networks, and higher per-capita commercial floor space — which is reflected in the domestic and commercial sectors growing faster than population alone would suggest. Industrialisation adds a second, compounding layer: as manufacturing and construction expand to employ a growing urban workforce, energy-intensive industries (steel, cement) pull coal and electricity demand up even faster than GDP growth alone, since these are among the most energy-intensive segments of the economy. Together, these forces explain why India's total energy consumption (3.05% CAGR) has grown faster than its population (1.11% CAGR) but roughly in line with its GDP (5.65% CAGR) once the modest gains in energy efficiency (energy intensity fell at -2.46% CAGR) are netted out.

3. India's Current Energy Mix and Fossil Fuel Dependence

3.1 Percentage Contribution of Each Energy Source

Table 8.11 of the report tracks the fuel share of major commodities in Total Primary Energy Supply (TPES). In FY 2022-23(P):

Energy Source	Share in TPES 2013-14 (%)	Share in TPES 2022-23(P) (%)
Coal	56.78	58.98
Crude Oil	39.42	31.48
Natural Gas	8.32	6.61
Nuclear	1.52	1.41
Renewable Energy	2.96	3.79

Source: Energy Statistics India 2024, Table 8.11. Columns need not sum to exactly 100% owing to rounding.

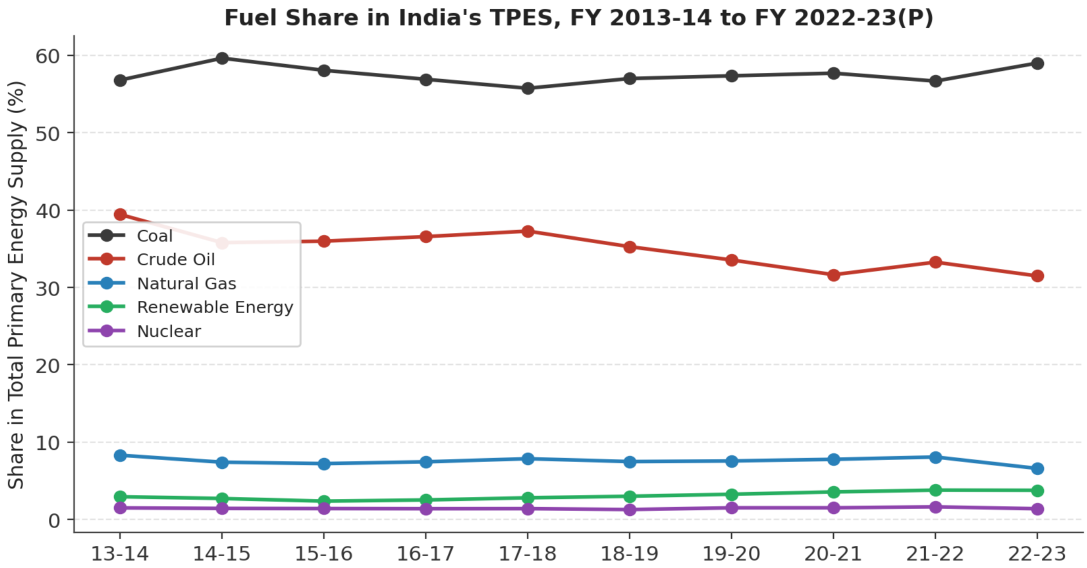

Figure 5: Fuel Share in India's Total Primary Energy Supply (TPES), FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 8.11.

Adding coal, crude oil and natural gas together, fossil fuels supplied roughly 97.1% of TPES in FY 2022-23(P), barely changed from about 97.2% a decade earlier — despite renewable energy's share nearly doubling in relative terms (2.96% to 3.79%). The picture is similar in Total Final Consumption (TFC, Table 8.12): oil products account for 41.18%, coal 30.55%, electricity 21.88% and natural gas 6.39% in FY 2022-23(P) — a mix that has shifted only modestly, mainly a swing from coal towards oil products and electricity.

3.2 Trends in Renewable Energy Adoption

Renewable adoption looks considerably more dynamic when viewed through the electricity sector specifically, rather than the whole primary-energy basket. Installed RES capacity (excluding large hydro) grew at 15.21% CAGR, more than double the growth of thermal capacity, taking the renewable-plus-hydro share of total electricity generation from 17.22% in FY 2013-14 to 20.34% in FY 2022-23(P) (Table 8.13/8.14). Within the electricity generation mix for FY 2022-23(P): thermal 77.18%, hydro (large) 8.81%, renewables (excl. hydro) 11.52%, and nuclear 2.49%.

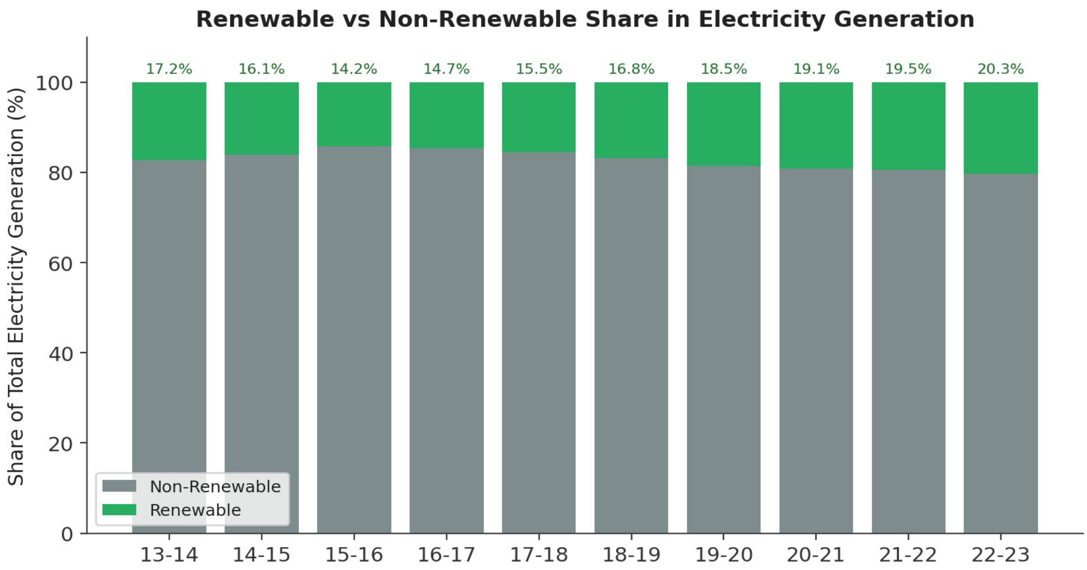

Figure 6: Renewable vs Non-Renewable Share in Total Electricity Generation, FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 8.14.

3.3 Evaluating Progress Toward a Cleaner Energy System

The report's data supports a nuanced conclusion. On the positive side, India has added renewable capacity far faster than fossil capacity for a decade running, driven down solar tariffs, and — per subsequent government and IEA reporting beyond FY 2022-23(P) — crossed 50% non-fossil installed generation capacity in 2025, five years ahead of its 2030 target, with cumulative renewable capacity reaching about 223.6 GW by mid-2025. On the less positive side, because renewables (particularly solar and wind) have much lower capacity-utilisation factors than coal plants, their fast capacity growth has translated into a much slower rise in their share of actual electricity generated, and an even smaller rise in their share of total primary energy supply. India's energy transition to date should therefore be read as 'renewables added on top of a still-growing fossil base' rather than fossil fuels being displaced — coal consumption itself grew 4.02% CAGR over the same decade renewable capacity was expanding at 15%+.

4. Per Capita Energy Consumption: India in Global Context

4.1 India's Domestic Per-Capita Trend

Table 8.2 of the report shows per-capita energy consumption rising from 21,419 megajoules (MJ) in FY 2013-14 to 25,424 MJ in FY 2022-23(P) — a modest 1.92% CAGR, well below the 3.05% growth in total consumption, because population itself grew throughout the period. Notably, energy intensity (energy consumed per unit of GDP) fell from 0.2737 to 0.2188 MJ per rupee — a CAGR of -2.46% — indicating that India is gradually decoupling economic growth from energy growth, i.e., producing more GDP per unit of energy consumed each year.

4.2 India vs. China, the United States and Germany

Independent international data (Energy Institute Statistical Review of World Energy, 2023 figures) place India's per-capita primary energy consumption at approximately 27.3 gigajoules (GJ) — a similar order of magnitude to the report's own 25,424 MJ (≈25.4 GJ) figure for FY 2022-23, the difference reflecting different measurement conventions and reference years. Against this benchmark, China consumes about 120 GJ per person (4.4 times India's level) and the United States about 277 GJ per person (over 10 times India's level); the world average stands at about 77 GJ per person, meaning India's per-capita consumption is roughly a third of the global average. Electricity consumption per capita tells a similar story: India's per-capita electricity use is well below 2,000 kWh per year, compared with roughly 6,000 kWh in Germany, over 7,000 kWh in China, and well over 12,000 kWh in the United States.

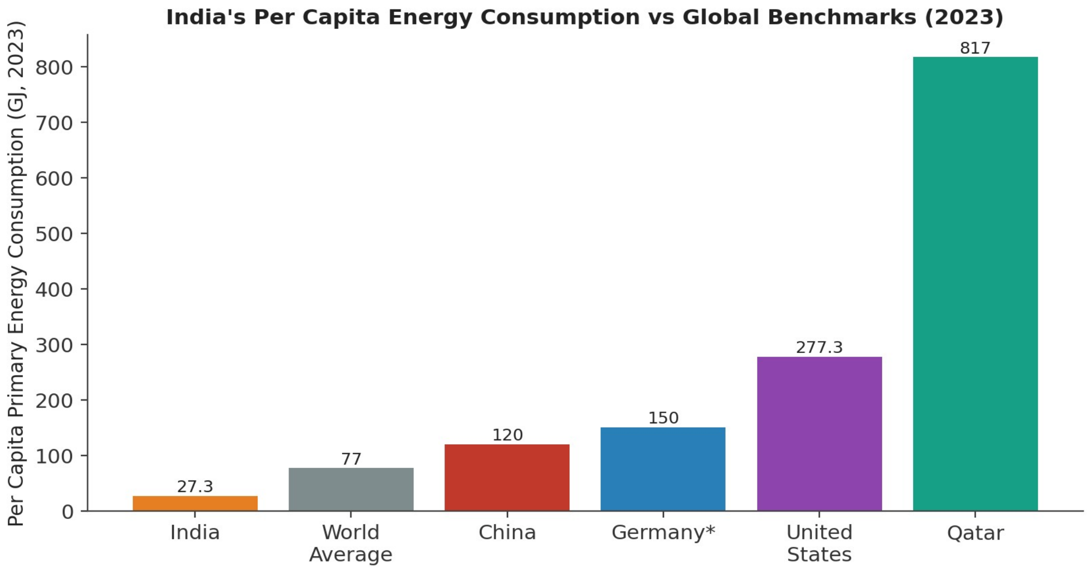

Figure 7: Per Capita Primary Energy Consumption, India vs Global Benchmarks, 2023. Sources: Energy Statistics India 2024 (Table 8.2, India domestic series); Energy Institute Statistical Review of World Energy 2024, as reported by ORF and Visual Capitalist (international comparison).

4.3 Explaining the Difference

- Stage of economic development: India's per-capita income remains well below China's and far below the US's, and energy consumption per person tends to rise with income and industrial output.
- Industrial structure: China's economy is far more manufacturing- and heavy-industry-intensive per capita than India's still-developing industrial base, driving much higher per-capita coal and electricity use.
- Population size and dilution effect: despite India having the world's third-largest absolute primary energy consumption (behind China and the US), its enormous population (over 1.4 billion) spreads that consumption very thin on a per-capita basis.

- Climate control and living standards: air-conditioning penetration, heating needs, and appliance ownership are all lower in India than in Germany or the US, holding down household energy use even as urban incomes rise.
- Infrastructure maturity: developed economies like Germany and the US have built out energy-intensive infrastructure (extensive vehicle fleets, larger housing stock, established heavy industry) over many decades that India is still constructing.

4.4 Implications for Future Demand

Because India's per-capita consumption is so far below global peers, and because rising incomes historically track with rising per-capita energy use, India's energy demand has substantial structural room — and pressure — to keep growing for decades even if population growth slows. The International Energy Agency has projected India will be the largest single source of global energy demand growth through 2035, driven precisely by this gap between current per-capita consumption and the levels associated with continued development.

5. Environmental Implications: Emissions and Climate Change

5.1 Emissions Associated with Different Energy Sources

Table 8.3 of the report (sourced from India's Third Biennial Update Report to the UNFCCC) breaks down energy-sector greenhouse gas emissions. Total energy-related emissions rose from about 1,651,928 Gg CO2-equivalent in 2011 to 2,129,428 Gg CO2-equivalent in 2016 (the latest year covered in this series). Within this, 'Energy Industries' (chiefly electricity and heat generation) is consistently the single largest source, contributing over 56% of fuel-combustion emissions throughout the period — a direct consequence of coal's dominant role in power generation. Manufacturing and construction is the second-largest contributor, followed by transport, whose emissions grew about 24% between 2011 and 2016 alone, faster than total emissions growth, reflecting rising vehicle numbers and freight movement.

GHG Source	2011 (Gg CO2e)	2016 (Gg CO2e)	Growth 2011-16 (%)
Energy Industries (power/heat)	9,24,258	12,06,587	30.5
Manufacturing & Construction	3,38,816	3,97,739	17.4
Transport	2,21,202	2,74,434	24.1
Other Sectors	1,20,228	2,13,490	77.6
Fugitive emissions (fuels)	47,426	37,179	-21.6
Total Energy Sector	16,51,928	21,29,428	28.9

Source: Energy Statistics India 2024, Table 8.3, citing India's Third Biennial Update Report to the UNFCCC (2021). This is the most recent officially compiled series in the source publication; more recent inventories exist but are outside the scope of this publication.

5.2 Sectors Contributing the Highest Emissions

Power generation (dominated by coal) is unambiguously the largest emitting activity in India's energy sector, consistent with coal's ~59% share of TPES and the fact that the power sector alone consumes over 70% of all coal used nationally. Manufacturing/construction and transport follow. Independent 2023-24 data corroborate this picture: India's coal consumption for power generation reportedly reached about 749 million tonnes in 2023 — more than the United States and Europe combined — even though India's per-capita coal use remains well below both regions, underscoring that India's emissions profile is driven primarily by the scale and coal-intensity of its power sector rather than by inefficient end-use.

5.3 Policies and Technologies Aimed at Reducing Emissions

- Panchamrit climate commitments (COP26, 2021): 500 GW of non-fossil electricity capacity by 2030, 50% of electricity capacity from non-fossil sources by 2030, a 45% reduction in the emissions intensity of GDP (from 2005 levels) by 2030, a cumulative 1 billion tonne reduction in projected CO2 emissions by 2030, and net-zero emissions by 2070.
- Renewable capacity expansion — solar, wind and battery storage — displacing some new coal capacity additions at the margin, even as absolute coal generation continues to rise to meet demand growth.
- Energy efficiency programmes (PAT, UJALA — discussed in detail in Section 6) that reduce emissions intensity per unit of output or per household.
- Ethanol blending in petrol, which reached a 20% blending rate by 2025, reducing crude-oil-linked transport emissions at the margin.
- A national carbon market for select energy-intensive industries, due to launch in 2026, intended to put an explicit price on carbon for large emitters.
- Nuclear capacity expansion, targeted to rise from about 8 GW today to 100 GW by 2047, as a low-carbon complement to renewables for baseload power.

6. Effectiveness of Government Energy Efficiency and Renewable Energy Initiatives

6.1 National Solar Mission

Launched in 2010 and scaled up in subsequent phases, the National Solar Mission (along with complementary schemes such as PM-KUSUM for agricultural solar pumps and PM Surya Ghar for rooftop solar, launched in February 2024) has driven a dramatic build-out of solar capacity: from about 2.8 GW in FY 2013-14 to roughly 108 GW by mid-2025. Solar tariffs fell by about 65% over the same period — from roughly Rs 6.17/kWh in FY 2014-15 to about Rs 2.15/kWh in FY 2024-25 — making Indian solar power among the cheapest in the world and self-sustaining without the same level of subsidy support the mission originally required.

6.2 Perform, Achieve and Trade (PAT) Scheme

PAT is a market-based mechanism that sets mandatory energy-efficiency targets for large energy-intensive industrial plants (steel, cement, fertiliser, textiles, thermal power, and others) and allows over-achievers to sell tradable energy-saving certificates to under-achievers. The first PAT cycle, completed in March 2015, delivered energy savings of about 8.67 million tonnes of oil equivalent — exceeding its original target — and the scheme has since been extended through multiple further cycles (currently in Cycle VII/VIIA), progressively widening its industrial coverage. This has directly supported the modest but real decline in India's energy intensity of GDP noted in Section 4.1.

6.3 UJALA (Unnat Jyoti by Affordable LEDs for All)

UJALA, launched in 2015 as a scale-up of the earlier Domestic Efficient Lighting Programme, has become the world's largest zero-subsidy LED distribution programme, having distributed well over 36 crore LED bulbs to households by early 2025 (with cumulative sales figures — including through allied retail channels — reported even higher). Its companion Street Lighting National Programme has installed over 1.3 crore LED streetlights nationwide, financed through an annuity model that required no upfront investment from municipalities, delivering substantial and durable electricity savings in the domestic and municipal-lighting segments.

6.4 Renewable Energy Targets

India met its 2030 target of 50% non-fossil-fuel electricity generation capacity in 2025 — five years ahead of schedule — with cumulative renewable capacity reaching about 223.6 GW by mid-2025, after a record single-year addition of about 29.5 GW in FY 2024-25. This is consistent with the report's own data showing RES capacity CAGR of over 15% through FY 2022-23(P), and confirms that the fast pace of renewable capacity addition identified in Section 3 has continued and even accelerated beyond the report's coverage period.

6.5 Assessing Overall Impact and Remaining Gaps

Taken together, these programmes have delivered clear, measurable wins: falling solar costs, hundreds of millions of efficient lighting units deployed, verified industrial energy savings under PAT, and an early achievement of the 2030 non-fossil capacity target. However, the report's own decade-long data point to several remaining gaps:

- Import dependency has risen, not fallen: net import dependency for crude oil grew from 83.35% in FY 2013-14 to 88.86% in FY 2022-23(P), and for natural gas from 33.46% to 43.30% over the same period (dipping mid-decade before climbing again) — despite the renewable energy push, indicating that energy-security goals and decarbonisation goals have not moved in tandem.

- Coal consumption keeps growing in absolute terms (4.02% CAGR), because policy has so far added renewable capacity on top of a still-expanding fossil base rather than substituting for it.
- Grid infrastructure and storage remain bottlenecks: transmission and distribution losses stood at 18-23% across the decade, and India's own projections call for over 200,000 km of new transmission lines and 230+ GWh of battery storage by 2030 to fully integrate variable renewables.
- PAT and UJALA address efficiency and lighting well but have limited reach into transport-sector fuel efficiency and into the fast-growing commercial/services sector's electricity use.

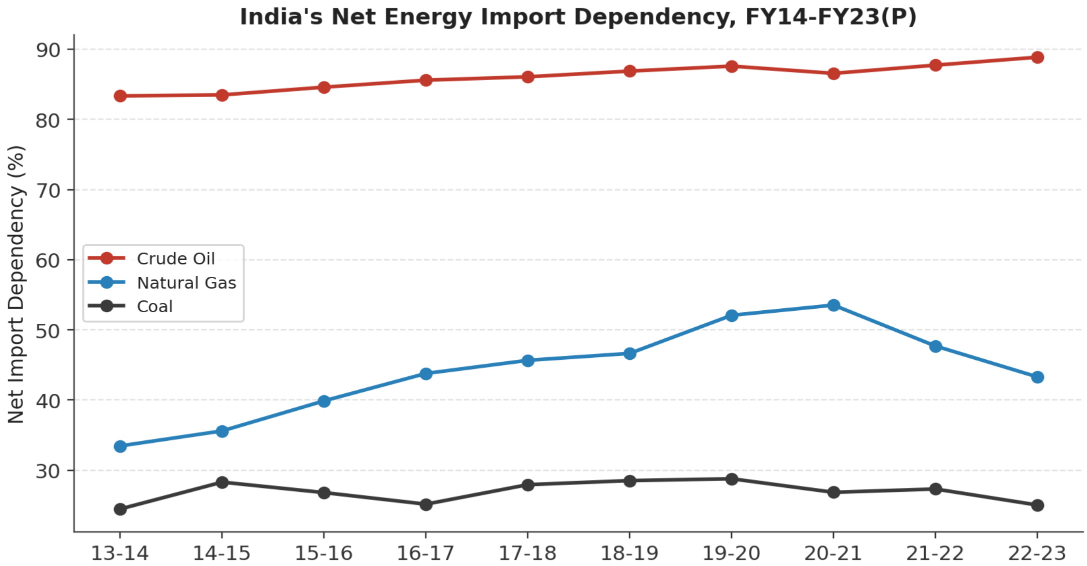

Figure 8: India's Net Energy Import Dependency, FY 2013-14 to FY 2022-23(P). Source: Energy Statistics India 2024, Table 8.15.

7. Future Energy Demand Scenarios and Recommendations

7.1 Forecast of Future Energy Demand

Extrapolating the report's own decade-long CAGR of 3.05% for total primary energy consumption would put India's demand at roughly 47,300 PJ by FY 2032-33 — but independent forecasts suggest even faster growth is likely as India's per-capita gap with global peers (Section 4) closes. The International Energy Agency's World Energy Outlook 2025 identifies India as the largest single source of global energy demand growth through 2035, projecting total energy demand to rise by more than 15 exajoules (roughly 15,000 PJ) by 2035 under its Stated Policies Scenario — growth comparable to adding the current energy demand of a large mid-sized economy. Electricity demand specifically is expected to nearly double, with power demand forecast to reach about 817 GW by 2030 (from roughly 425 GW of installed capacity in FY 2022-23(P)), and coal demand is expected to keep rising moderately through 2035 even as its share of the mix declines, because it continues to provide dispatchable, flexible baseload generation that variable renewables cannot yet fully replace.

7.2 Key Challenges: Energy Security, Import Dependence and Infrastructure

- Energy security: with crude oil import dependency near 89% and rising, and natural gas import dependency above 43%, India remains highly exposed to global price volatility and supply disruptions — a vulnerability the government is addressing partly by diversifying crude oil sourcing from 27 to over 40 countries.
- Grid integration: rapidly growing variable renewable capacity requires large investments in transmission (over 200,000 km of new lines planned, including 60,000 km dedicated to renewable evacuation) and storage (230+ GWh of battery storage targeted by 2030) to avoid curtailment and maintain reliability.
- Distribution losses: aggregate technical and commercial (AT&C) losses remain high (over 21% in recent estimates), representing both a financial drain on power utilities and wasted generation capacity.
- Capital intensity: meeting the 500 GW non-fossil capacity target and associated grid upgrades requires sustained, large-scale investment, including in domestic manufacturing of solar modules and batteries.

7.3 Recommended Strategies for Sustainable and Secure Energy Supply

- Accelerate renewable capacity addition beyond current targets, with continued focus on solar, wind and, increasingly, offshore wind and green hydrogen, to keep pace with — and eventually outpace — fossil-fuel demand growth rather than merely supplementing it.
- Scale up battery and pumped-hydro storage in line with the ~230 GWh 2030 target, to firm up variable renewable output and enable a higher share of renewables in actual generation, not just installed capacity.
- Expand and modernise transmission infrastructure, including dedicated renewable-energy corridors, and invest in smart grid and demand-response technologies to manage a more variable, distributed generation mix.
- Broaden and deepen energy-efficiency programmes (PAT, UJALA, ECBC for buildings) to extend beyond current industrial and lighting coverage into transport fuel efficiency, commercial building codes, and appliance standards, reinforcing the energy-intensity decline already visible in the data.
- Diversify energy imports and accelerate domestic exploration and production to manage rising oil and gas import dependency, while expanding ethanol blending and electric mobility to structurally reduce oil demand in transport over time.
- Scale up nuclear capacity toward the 100 GW-by-2047 target to provide low-carbon baseload power that complements variable renewables.

- Pursue demand-side management in agriculture (efficient, possibly solar-powered irrigation pumps under PM-KUSUM) and in the fast-growing domestic/commercial sectors (efficient appliances, building codes) to moderate demand growth without constraining development.
- Prepare for a genuinely decoupled growth path — where GDP keeps growing while emissions intensity keeps falling — by tracking energy-intensity and emissions-intensity indicators (already declining at -2.46% and improving per the report) as the primary yardsticks of policy success, rather than capacity additions alone.

8. Conclusion

Over the past decade, India's energy story has been one of strong, broad-based growth: total consumption rising at 3.05% CAGR, sectoral demand climbing fastest in the domestic and agricultural segments, and renewable capacity expanding more than three times faster than the system as a whole. Yet the underlying energy mix has shifted only modestly — coal, oil and gas still supply roughly 97% of primary energy, import dependency for oil and gas has risen rather than fallen, and India's per-capita consumption remains a fraction of global peers, leaving substantial room — and pressure — for continued demand growth. Government programmes like the National Solar Mission, PAT and UJALA have delivered genuine, measurable progress, including an early achievement of the 2030 non-fossil capacity target, but the scale of India's development needs means that securing a reliable, affordable and genuinely low-carbon energy future will require renewables and efficiency gains to grow even faster than they have so far — fast enough not just to add to the system, but to begin displacing fossil fuels within it.

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